

ZX Spectrum 48K → PC Keyboard Map

All 40 Spectrum keys map one-to-one onto a standard 104/105-key PC keyboard. Legends below are extracted from the 48K ROM's own key tables, so they match what the machine actually produces — not what is silk-screened on any particular case.

THE SPECTRUM KEYBOARD

EDIT		CAPS LOCK		TRUE VIDEO		INV VIDEO		←		↓		↑		→		GRAPHICS		DELETE	
1 PAPER DEF FN PLOT	2 FLASH FN DRAW	3 BRIGHT LINE REM	4 INVERSE OPEN # RUN	5 OVER CLOSE # RANDOMIZE	6 AT MOVE RETURN	7 TAB ERASE IF	8 POINT INPUT	9 CAT POKE	0 INK FORMAT PRINT										
Q SIN ASN NEW	W COS ACS SAVE	E TAN ATN DIM	R INT VERIFY FOR	T RND MERGE GO TO	Y STR\$ [GO SUB	U CHR\$] LOAD	I CODE IN LIST	O PEEK OUT LET	P TAB ©										
A READ ~	S RESTORE COPY	D DATA \ CLEAR	F SGN { CONTINUE	G ABS } CLS	H SQR CIRCLE BORDER	J VAL VAL\$ NEXT	K LEN SCREEN\$ PAUSE	L USR ATTR	ENTER										
CAPS SHIFT		Z LN BEEP	X EXP INK	C LPRINT PAPER	V LLIST FLASH	B BIN BRIGHT	N INKEY\$ OVER	M PI INVERSE	SYM SHIFT		SPACE				BREAK				

Above the key — BASIC keyword (K mode), or CAPS SHIFT function on the digits On the key, red — SYM SHIFT Below, green — EXTENDED mode Below, red — EXTENDED + SYM SHIFT

THE SAME KEYS ON A STANDARD PC KEYBOARD

		1	2	3	4	5	6	7	8	9	0	-	=	Backspace		
Tab		Q	W	E	R	T	Y	U	I	O	P	[	]	\		
Caps Lock		A	S	D	F	G	H	J	K	L	;	'	Enter			
Shift CAPS SHIFT		Z	X	C	V	B	N	M	,	.	/	Shift SYM SHIFT				
Ctrl	Win	Alt	Space										Alt	Win	Menu	Ctrl

Mapped — wired in keyb_lookup.h today Left Shift = CAPS SHIFT, Right Shift = SYM SHIFT Combination key — exists on the Spectrum only as SHIFT + key (see overleaf) No equivalent

Matrix Wiring & Shifted Combinations

The keyboard is an 8×5 matrix read through the ULA. Any port with bit 0 clear selects it; the high byte of the port chooses which rows to scan. Rows are active *low*, and several may be selected at once, in which case their results are ANDed — that is how the ROM detects a shifted key. A pressed key reads back as **0**.

MATRIX MAP

ROW	ADDRESS LINE	PORT	BIT 0	BIT 1	BIT 2	BIT 3	BIT 4
0	A8	0xFEFE	CAPS SHIFT Left Shift	Z	X	C	V
1	A9	0xFDFE	A	S	D	F	G
2	A10	0xFBFE	Q	W	E	R	T
3	A11	0xF7FE	1	2	3	4	5
4	A12	0xEFFE	0	9	8	7	6
5	A13	0xDFFE	P	O	I	U	Y
6	A14	0xBFFE	ENTER Enter	L	K	J	H
7	A15	0x7FFE	SPACE Space	SYM SHIFT Right Shift	M	N	B

Rows 3 and 4 run in opposite directions. Row 3 counts 1 2 3 4 5 outward from the left edge of the keyboard and row 4 counts 0 9 8 7 6 outward from the right, mirroring the physical wiring. This is a classic source of transposed-key bugs.

SHIFTED COMBINATIONS A PC USER EXPECTS

The Spectrum has no dedicated Backspace, cursor, comma or period keys — they are all SHIFT combinations. To support the PC keys below, one scancode must pull *two* matrix bits low.

PC KEY	CAPS SHIFT +	MATRIX BITS	PRODUCES	CHAR	SYM SHIFT +	MATRIX BITS	CHAR	SYM SHIFT +	MATRIX BITS
Backspace	0	(0,0) + (4,0)	DELETE	,	N	(7,1) + (7,3)	/	V	(7,1) + (0,4)
← Left	5	(0,0) + (3,4)	cursor left	.	M	(7,1) + (7,2)	*	B	(7,1) + (7,4)
↓ Down	6	(0,0) + (4,4)	cursor down	;	O	(7,1) + (5,1)	?	C	(7,1) + (0,3)
↑ Up	7	(0,0) + (4,3)	cursor up	'	7	(7,1) + (4,3)	:	Z	(7,1) + (0,1)
→ Right	8	(0,0) + (4,2)	cursor right	"	P	(7,1) + (5,0)	@	2	(7,1) + (3,1)
Caps Lock	2	(0,0) + (3,1)	CAPS LOCK	-	J	(7,1) + (6,3)	£	X	(7,1) + (0,2)
Ctrl (suggested)	SPACE	(0,0) + (7,0)	BREAK	=	L	(7,1) + (6,1)	<	R	(7,1) + (2,3)
				+	K	(7,1) + (6,2)	>	T	(7,1) + (2,4)

Extended mode reaches the digit keywords only with SYM SHIFT. EXTENDED then a *letter* gives that letter's keyword directly, but EXTENDED then a *digit* gives an embedded control code (INK, PAPER, FLASH, BRIGHT, INVERSE, OVER, AT, TAB) — the ROM computes these with SUB \$20 at 0x037B rather than tabulating them. DEF FN , FN , LINE and the rest need EXTENDED + SYM SHIFT + digit. Enter extended mode with both shifts together (Left Shift + Right Shift); it lasts for one keypress.

This is a positional map, not a character map. Shift + 2 prints @ on a PC, but CAPS SHIFT + 2 on a Spectrum is CAPS LOCK — @ is SYM SHIFT + 2. Mapping physical positions keeps the emulator faithful; a "PC-friendly" typing mode is a separate translation layer.

Legends extracted from 48.rom: token table 0x0095; SYM SHIFT letters 0x026A; EXTENDED letters 0x022C; EXTENDED+SYM letters 0x0246; CAPS SHIFT digits 0x0260; EXTENDED + SYM digits 0x0284. K-mode keywords are token 0xE6 + (letter - 'A') and EXTENDED digit control codes are SUB \$20 / SUB \$36 at 0x037B, both computed rather than tabulated. SYM SHIFT digit symbols and every extended-mode path verified by running the ROM and reading LAST-K (0x5C08). Matrix assignments mirror include/z80/keyb_lookup.h.